batplot

User Manual

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Interactive CLI Plotting Tool for Battery & Materials Characterization Data

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https://github.com/chem-plot/batplot

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1 What Is *batplot*?

batplot is a lightweight, open-source Python command-line interface (CLI) tool designed for rapid, publication-quality visualization of battery and materials characterization data. It reads raw instrument output files directly and produces fully styled, interactive figures from a single terminal command, with no scripting required.

Developed by a battery materials research group, *batplot* was born out of genuine visualization needs for 1D (XY) plots, electrochemical cycling, and combined *operando* / *in situ* 2D contour plots, with all three under customized intuitive interactive styling interface.

Wether you use Python or commercial softwares such as Origin for your plotting, *batplot* can speed up the data visualization process and help you shift effort to scientific interpretation.

Key capabilities:

- Zero scripting required. Every workflow runs from a single terminal command.
- Interactive menus: live, code-free customization of colors, geometries, spines, cycles and many more.
- Session save: save incomplete figure projects and reload them later.
- Style export: export visual configuration files.
- Batch processing: apply a style file to every matching file in a folder.

Supported Data Types

batplot handles all major data types encountered in modern battery materials research:

Data Category	Supported Formats
`XRD / PDF / XAS`	.xye, .xy, .qye, .dat, .csv, .txt, .brml (Bruker), .raw (Bruker), .gr, .nor
`Crystallography`	.cif
`Electrochemistry (Neware)`	.csv/.xlsx (GC, dQdV, CPC, EPC)
`Electrochemistry (Biologic)`	.mpt (GC, CV, CPC, EPC)
`Generic two-column data`	Any text/Excel file

Three Interactive Plotting Modes

batplot operates in three primary interacitve modes which can be activated using simply one line of command, *batplot* reads the data files and plot them with sensible default, each tailored to a different class of experiment:

- 1D / XY Mode: plot 2-column data. Optimized for XRD with Q / 2θ conversion and CIF tick overlays, but also support other types of text/Excel data files.

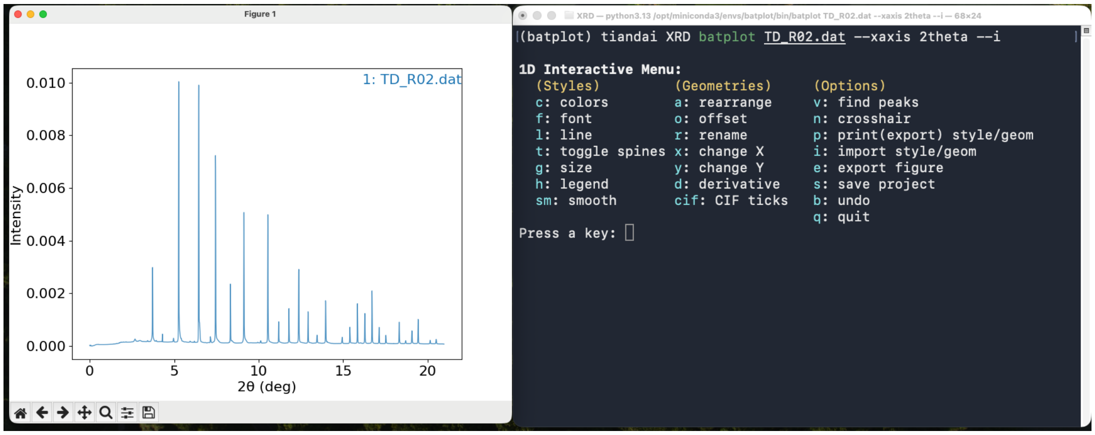

Figure 1: 1D mode

- Electrochemistry (EC) Mode: galvanostatic cycling (GC), cyclic voltammetry (CV), differential capacity (dQ/dV), and capacity/energy-per-cycle (CPC/EPC) plots from Biologic and Neware output files.

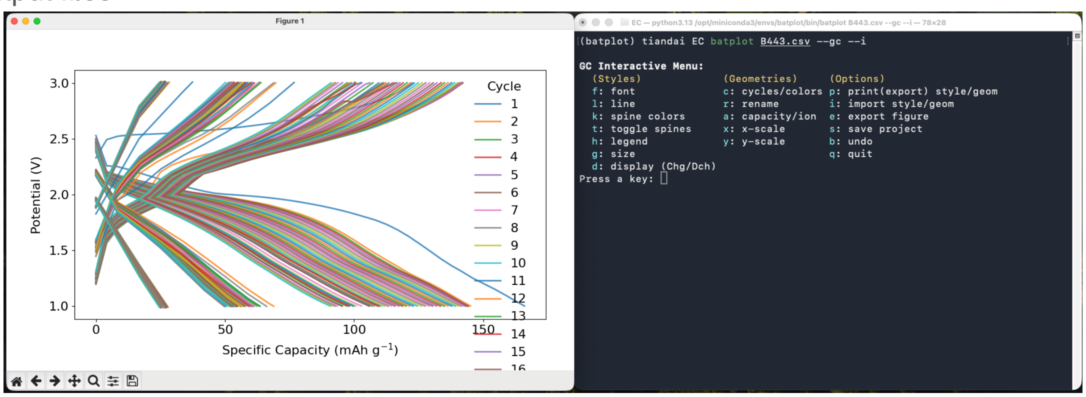

Figure 2: EC (GC) mode

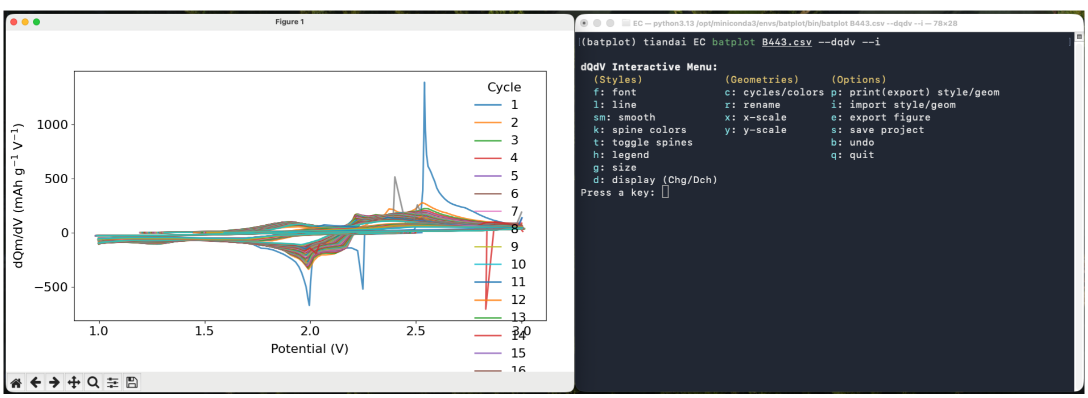

Figure 3: EC (dQ/dV) mode

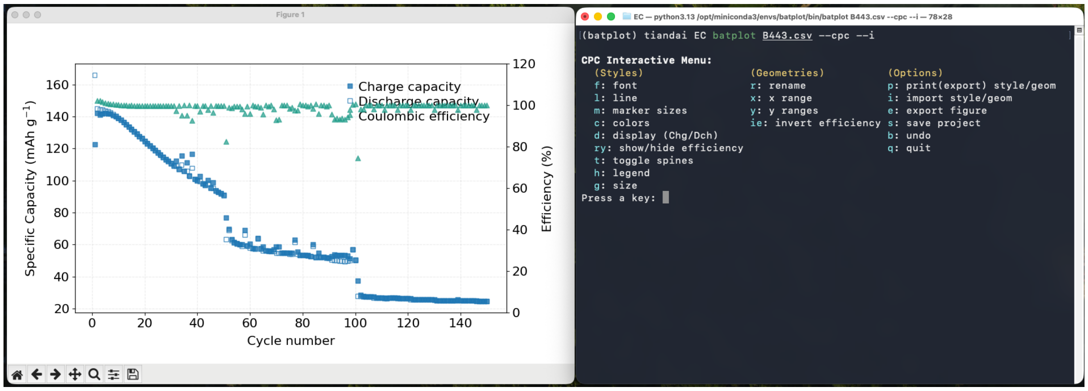

Figure 4: EC (CPC) mode

- *Operando*/Contour Mode: assembles a folder of sequential files into a 2D contour map and automatically plot electrochemical data as a synchronized side panel (if available).

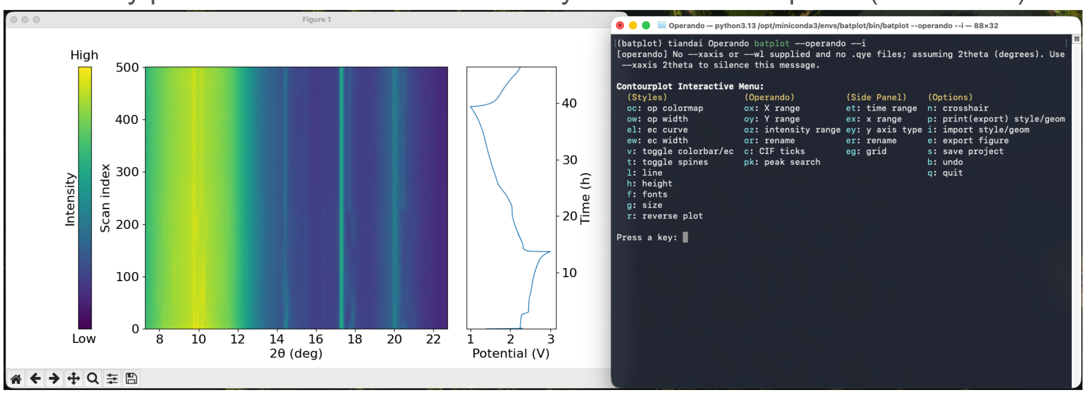

Figure 5: *Operando* mode

2 Installation

Anaconda Prompt is needed prior to using *batplot*. Anaconda is a distribution platform for Python that manages different Python versions and packages. The key tool is a command called `conda`, which is a package and environment manager.

batplot relies on packages such as *matplotlib* and *numpy* for its functionality. These packages come in different versions that you may already have that are not compatible with *batplot*. By using `conda`, you can create an isolated environment so *batplot* installs the exact versions it needs.

Step 1: Download Anaconda Prompt

Download and install Anaconda from the official Anaconda website:
https://www.anaconda.com/download

Once installation is complete, search for a program called Anaconda Prompt. You will see a terminal window with (base) shown on the far left of the prompt line, indicating the default conda environment.

Step 2: Create a dedicated batplot environment

In the Anaconda Prompt, type the following command:

```
conda create -n batplot python=3.13
```

conda will resolve the environment and ask you to confirm. Type Y and press Enter to proceed. This creates a new isolated Python 3.13 environment named batplot.

> **Note:** On some systems the version number must be quoted:
> `conda create -n batplot python="3.13"`

Step 3: Activate the batplot environment

Activate the newly created environment:

```
conda activate batplot
```

The label on the far left of the prompt will change from (base) to (batplot), confirming that the environment is now active.

Step 4: Install batplot

Install batplot from PyPI:

```
pip install batplot
```

pip will automatically download batplot and all required dependencies (*matplotlib*, *numpy*, etc.) into the batplot environment.

Step 5: Verify the installation

Confirm that batplot installed correctly by running:

```
batplot --help
```

After a few seconds a help message will appear. At the top of the message you should see the current version number (e.g., v1.8.29). If you see this, the installation was successful.

***batplot* only works inside its conda environment. Each time before using, open Anaconda Prompt and activate the environment first:**

```
conda activate batplot
```

> **Note:** batplot is actively developed. New features are added regularly.

3. Basics of using terminal

The terminal (using Anaconda Prompt) is a text-based interface where you type commands to control your computer. *batplot* runs entirely from the terminal, so knowing a few basic navigation commands is all you need to get started. The sections below cover the essentials.

Seeing Your Current Location: `pwd`

If you ever lose track of where you are, type `pwd` (print working directory) and press Enter. The terminal prints the full path of your current folder.

```
pwd
```

Navigating Between Folders: `cd`

The `cd` command (change directory) moves you from one folder to another. Type `cd` followed by the path of the folder you want to enter, then press Enter. The prompt updates to confirm your new location. If the folder name contains spaces, enclose the entire path in quotation marks.

```
# Enter a subfolder called Data
cd Data
# Go up one level to the parent folder
cd ..
# Jump directly to a folder using its full path
cd C:\Users\YourName\Documents\XRD_Data
batplot xrd.xy -xaxis 2theta -I # Using batplot to plot an XRD data file
```

The shorthand `..` always means the parent folder (one level up). Chaining it — for example `cd ..\..` — moves up two levels. A full (absolute) path like the example above works regardless of where you currently are.

Listing Files in a Folder: `ls / dir`

To see the files and folders inside your current directory, use `ls` on macOS / Linux, or `dir` on Windows Anaconda Prompt. This is useful for confirming that your data files are present before running a batplot command.

```
# macOS / Linux
ls
# Windows Anaconda Prompt
dir
```

Tab Completion

After typing the first few characters of a name, press the `Tab` key and the terminal will complete the name automatically. If more than one match exists, pressing `Tab` again cycles through the options.

Recalling Previous Commands: Arrow Keys

Press the up arrow key to scroll back through previously entered commands. This is especially useful when you want to repeat a batplot command with a small change, press the up arrow, edit the line, and press Enter to run it again.

> **Tip:** On Windows, the quickest way to navigate to a data folder is to open it in File Explorer, click the address bar to select the path, copy it, and then paste it into Anaconda Prompt after `cd`. On macOS, you can drag a folder from Finder directly into the Terminal window to insert its full path automatically.

4 How to Use *batplot*

General Command Structure

batplot command follows this pattern:

```
batplot  [path/]/[files...]/[keyword]  [--flags]
```

Component	Description
`path/`	Optional. A folder path when data is not in the current directory. *batplot* looks for supported files in that folder.
`files...`	One or more data files to overlay.
`keywords`	Special keywords: allfiles (all supported files), allxyfiles (only .xy), allnorfiles (only .nor), etc.
`--flags`	Options that control mode, axes, styling, and output. All flags use the double-dash (`--`) prefix.

Getting Help

To see *batplot* help and documentation:

```
batplot --help            # General help
```

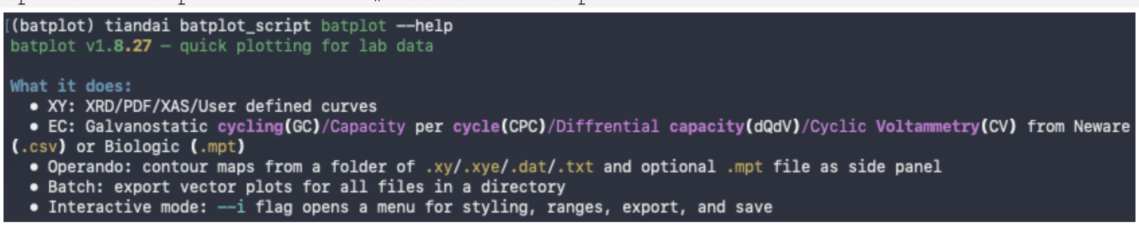

Figure 6: Help message

```
batplot --help xy         # 1D / XY mode help
```

```
batplot --help ec         # Electrochemistry mode help
```

```
batplot --help op         # Operando mode help
```

```
batplot --version         # Version and release notes
```

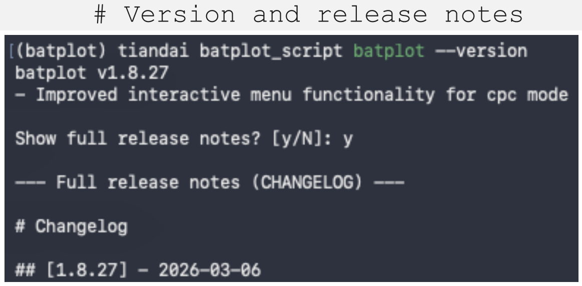

Figure 7: Version check

```
batplot --manual          # Open the illustrated manual
```

Recommended Workflow

As navigating to different paths can sometimes be tedious, you can create a folder named “Figures”, and use `cd` to navigate to this “Figures” path. Then, you can directly drag the files/copy the path into the terminal for interactive plotting. Inside interactive menu, you can easily save the plot session/export the figure/export the style/import a style from the “Figures” folder as batplot automatically detects your terminal path and use that to save or find relavant files. For example:

```
batplot drag/the/file.xy --xaxis 2theta -i
```

```
batplot drag/the/path --operando -i
```

Selecting Modes

The plotting mode is determined by which mode flag you include:

Mode	Flag(s) to add
`1D / XY (default)`	indicate data type by `--xaxis` flag
`Galvanostatic cycling`	`--gc`
`Cyclic voltammetry`	`--cv`
`Differential capacity`	`--dqdv`
`Capacity/Energy density per cycle`	`--cpc/--epc`
`Operando / contour`	`--operando` or `--contour`

The Interactive Menus

Customized interactive menus are the central feature that makes *batplot* unique. Add **`--i`** to any command to open a text-driven menu in the terminal alongside the live figure window. Every change you make, from colors and fonts to axis labels, line widths, and tick spacing, updates the figure in real time. The interactive menus are intuitive and are not included in this manual.

```
batplot file.txt --xaxis temp --i
```

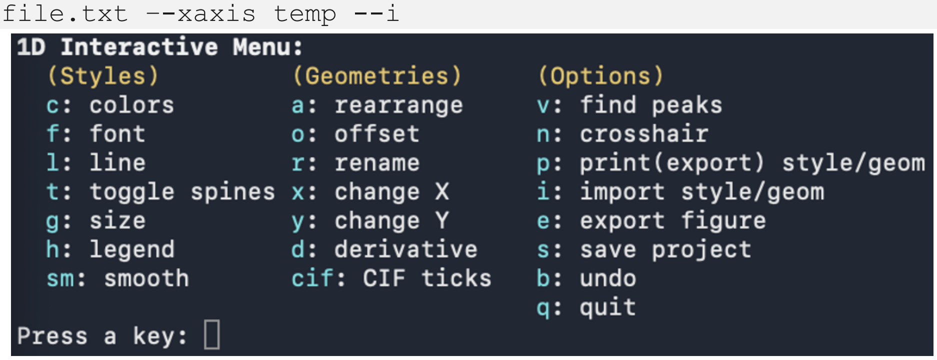

Figure 8: 1D mode

```
batplot file.csv --gc --i
```

```
GC Interactive Menu:
  (Styles)                (Geometries)      (Options)
  f: font                 c: cycles/colors p: print(export) style/geom
  l: line                 r: rename         i: import style/geom
  k: spine colors         a: capacity/ion   e: export figure
  t: toggle spines        x: x-scale        s: save project
  h: legend               y: y-scale        b: undo
  g: size                                   q: quit
  d: display (Chg/Dch)
Press a key:
```

Figure 9: GC mode

```
batplot file.csv --dqdv --i
```

```
dQdV Interactive Menu:
  (Styles)                (Geometries)      (Options)
  f: font                 c: cycles/colors p: print(export) style/geom
  l: line                 r: rename         i: import style/geom
  sm: smooth              x: x-scale        e: export figure
  k: spine colors         y: y-scale        s: save project
  t: toggle spines                          b: undo
  h: legend                                 q: quit
  g: size
  d: display (Chg/Dch)
Press a key:
```

Figure 10: dQ/dV mode

```
batplot file.csv --cpc --i
```

```
CPC Interactive Menu:
  (Styles)                  (Geometries)          (Options)
  f: font                   r: rename             p: print(export) style/geom
  l: line                   x: x range            i: import style/geom
  m: marker sizes           y: y ranges           e: export figure
  c: colors                 ie: invert efficiency s: save project
  d: display (Chg/Dch)                            b: undo
  ry: show/hide efficiency                        q: quit
  t: toggle spines
  h: legend
  g: size
Press a key:
```

Figure 11: CPC mode

```
batplot --operando -i
batplot --contour --i
```

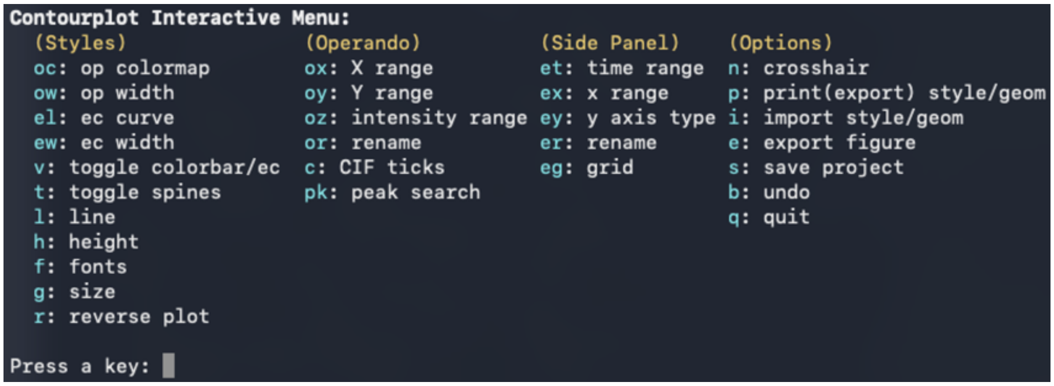

Figure 12: *Operando*/contour mode

Inside interactive menus, you can work on customized plot styling and carry out some basic data analysis, and you will also be able to:

- Press `p` to export the current style and `i` to import a style configuration file. You can either export .bps (only style) or .bpsg (style+geometries) and import the file.
- Press `s` to save the entire session (data + style) as a .pkl file for later resumption.
- Press `e` to export the Figure.

> **Note:** By default, style files (.bps, .bpsg) and image files (.svg, .png, etc.) are exported into a subfolder within the selected path, while session files (.pkl) are exported directly to the selected path.
>
> When using `i` to import a style file, *batplot* automatically scans the style subfolder in the current path, lists the detected files with numbers, and allows you to select a file by typing its corresponding number.

Download Tutorial Files

Below sections are some examples of how to use *batplot*, you can find the tutorial files in this link.

5 Examples: 1D Mode

The 1D / XY mode is the default mode of *batplot*. It is optimized for XRD but **support any generic two-column data. By default, *batplot* uses the first and second column to plot as x and y, this can be changed using `--readcol` flag.**

Example 5.0: Plot Any Two-column Data

```
batplot data.txt --xaxis Something --i
# plots with user defined X-axis
```

> **Note:** `--xaxis` flag is needed in most cases for 1D mode as *batplot* uses it to detect data type.

Example 5.1: Plot XRD Data with Wavelength Conversion

```
batplot pattern.xy --xaxis 2theta --i
# plots XRD data in 2theta space
```

```
batplot data.qye --xaxis q --i
# plots XRD data in q space
```

Tutorial

```
# Navigate to XRD folder
batplot TD_S62-64.xy --wl 1.54 --i
# converts 2θ → Q and plot in q space
```

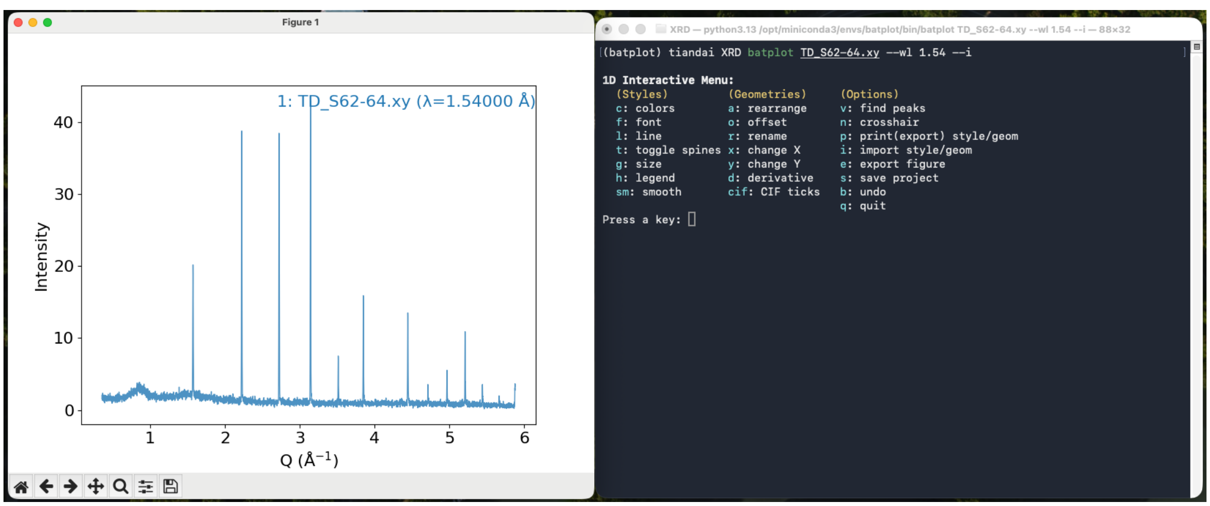

Figure 13: 1D mode

```
batplot data.xye:0.709 --i
# another way to convert 2θ → Q and plot
```

```
batplot data1.xye data2.txt --wl 1.54 --i
# multi-file overlay with same wavelength in q space
```

```
batplot data1.xye:0.709 data2.txt:1.54 --i
# multi-file overlay with different wavelengths in q space
```

```
batplot data.xye:0.709:1.54 --xaxis 2theta --i
# plots XRD data in 2theta with a different wavelength (0.709 → 1.54)
```

```
batplot file1.xy file2.xy --norm --i
# plots normalized data between the scale of 0-1
```

Wavelength Handling

batplot provides flexible wavelength handling for XRD data. Q and q are always equivalent (case-insensitive).

Syntax	Meaning
`--wl 1.5406`	Convert all 2θ data to Q using wavelength 1.5406 Å (Cu Kα)
`file.xye:1.54`	Per-file wavelength: convert this file's 2θ to Q using 1.54 Å
`file.xye:0.25:1.54`	Dual wavelength: re-project from λ_1=0.25 Å frame to λ_2=1.54 Å
`phase.cif:1.54`	Calculate CIF tick positions using 1.54 Å (for 2θ axis)

> **Note:** When `--wl` or `:wavelength` is provided, *batplot* converts the x-axis to Q-space and you do not need to add `--xaxis q`.

Example 5.2: Using `--readcol` to Specify Columns

```
batplot data.txt --readcol 2 4 --i
# plots column 2 and 4 as x and y
```

```
batplot data.txt --readcol 1 2 1 3 1 4 --i
# plots 3 curves with column 1 as x and column 2, 3, 4 as y
```

```
batplot data.txt --readcol 1 2-6 --i
# plots 5 curves with column 1 as x and column 2, 3, 4, 5, 6 as y
```

```
batplot data1.txt --readcol 1 2 data2.txt --readcol 2 4 --i
# plots 2 curves with specified columns for each
```

Example 5.3: Stacking Multiple Files Using `--stack`

```
batplot file1.xye file2.xye --xaxis 2theta --stack --i
```

```
batplot (your/optional/path) allfiles --stack --wl 1.54 -i
# with allfiles keyword, batplot reads every text/Excel file in the
directory, sorts them in natural order (scan2 before scan10), converts
them to Q-space using the per-file wavelength, and stacks them vertically.
```

Tutorial

```
batplot (path/to/XRD) alldatfiles --stack --xaxis 2theta --i # inside any
path

batplot alldatfiles --stack --xaxis 2theta --i # inside XRD folder

# plots all files with .dat extension in stack mode, you can use
all<ext>files keyword to plot all files with the specified extension.
```

Figure 14: 1D stack mode

Example 5.4: Plot with CIF Ticks

```
batplot data.xye phase.cif --wl 1.54 --i
# plots XRD data with CIF ticks in Q space
```

```
batplot data.xye phase.cif:1.54 --xaxis 2theta --i
# plots XRD data with CIF ticks in 2theta space
```

Tutorial

```
# Navigate to XRD folder
batplot TD_S62-64.xy:1.54 TD_R02.dat:0.25995 Li2FeSeO.cif Li2Se.cif --
stack --i
# stacks two XRD data files with two CIF ticks in Q space
```

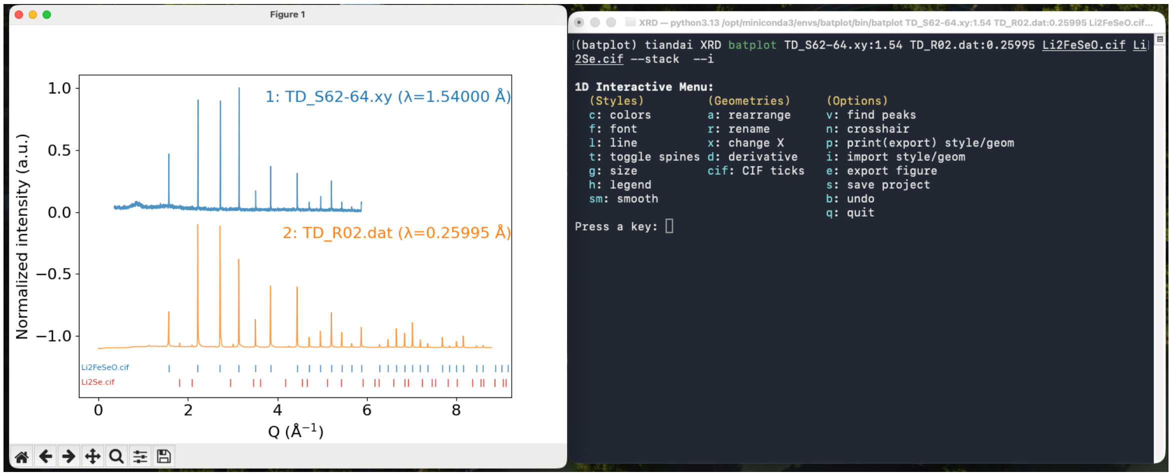

Figure 15: 1D stack mode with CIF ticks

Example 5.5: Derivative Plots

The `--1d` (and `--2d`) flag plots the first derivative dy/dx of each dataset. This is useful for identifying peak positions and inflection points. The derivative is computed using numpy's gradient function, which handles non-uniform x-spacing automatically.

```
batplot file1.nor file2.xy --1d --stack --i
```

```
batplot allfiles --2d --i
```

```
batplot file.nor --1d --xrange 10 80
```

Example 5.6: EXAFS k-Weighting

For EXAFS data in k-space, k-weighting options are available:

Flag	Transformation / use case
`--chik`	χ(k): standard oscillations
`--kchik`	k × χ(k): emphasize mid-k features
`--k2chik`	k^2 × χ(k): most common weighting, balances signal
`--k3chik`	k^3 × χ(k): emphasize high-k and heavy backscatterers

```
batplot data.txt --k2chik --i
```

```
batplot file1.chik file2.chik --k2chik --stack --i
```

Example 5.7: Batch Export

Export every supported file in the current folder as a separate SVG figure:

```
batplot --all --xaxis 2theta
batplot --all mystyle.bpsg      # apply saved style+geometry to every file
```

> **Note:** Adding a style file to apply it works for normal XY plot as well, it is the same as using `i` command in the interactive menu.
>
> `---all` flag is not the same as `allfiles` keyword

Output files are saved to the batplot_svg/ subdirectory automatically created in the current folder.

Example 5.8: Dual Axes Mode

Using `--ry` flag to plot files using right y axis:

```
batplot file1.xy file2.dat --ry --xaxis 2theta
# plots file2.dat against right y
```

```
Batplot file1.xy file2.xy --ry file3.xy --ry --xaxis 2theta
# multi-file support for right y axis
```

Example 5.9: Convert and Export XRD Data Files

Apart from plotting, batplot can convert XRD data freely between Q and 2 theta space under any wavelength and export the data files in a subfolder under the current path.

```
batplot data.xy --convert 1.54 q
# exports the data file under q space
```

```
batplot data.txt --convert 1.54 0.709
# exports the data file under a new wavelength
```

```
batplot data.dat readcol 2 4 --convert 1.54 0.709
# exports the data file under a new wavelength with selected x and y
```

```
batplot allfiles --convert 1.54 q
# converts and exports allfiles with given wavelength/q
```

6 Examples: Electrochemistry (EC) and CPC / EPC Modes

Electrochemistry mode supports four visualization types: galvanostatic cycling (GC), cyclic voltammetry (CV), differential capacity (dQ/dV), and capacity/energy density-per-cycle (CPC). Both Biologic (.mpt) and Neware (.csv) files are read directly. More instruments are supported upon request.

Requirement for data exporting from instruments:

Neware: Customizer report – check all boxes – export to `.csv` or `.xlsx` format

BioLogic: Export all to `.mpt` file

> **Note:** For BioLogic `.mpt` files in GC and CPC modes, the active material mass must be provided with `--mass` (in milligrams). For Neware `.csv`/`.xlsx` files, capacity data is already recorded in the file and `--mass` is not required.

Example 6.1: Galvanostatic Cycling (GC)

Plot potential vs. specific capacity charge/discharge profiles:

```
batplot battery.csv --gc --i
# from a Neware .csv file
```

```
batplot battery.mpt --gc --mass 7.0 --i
# from a Biologic .mpt file (mass required in milligrams)
```

```
batplot battery.mpt --gc --mass 12g --i
# from a Biologic .mpt file with high mass loading in grams
```

```
batplot BatA.csv BatB.csv BatC.csv --gc -i
# plots all three files
```

Tutorial

```
# Navigate to EC folder
batplot B443.csv --gc --i
```

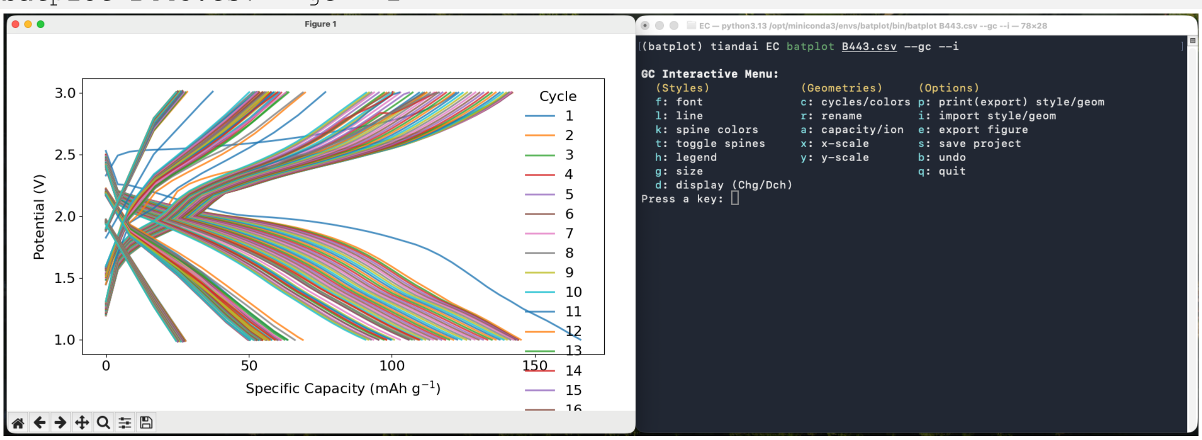

Figure 16: EC mode

Example 6.2: Cyclic Voltammetry (CV)

Plot current vs. voltage cycles for electrochemical characterization:

```
batplot cyclic.mpt --cv --i
```

Example 6.3: Differential Capacity (dQ/dV)

Plot dQ/dV vs. potential to identify electrochemical reaction peaks:

```
batplot battery.csv --dqdv –i
```

```
batplot 1.csv 2.csv --dqdv --i
# plots multiple files and compare the peaks
```

Example 6.4: Capacity Per Cycle (CPC)

CPC mode plots charge and discharge specific capacity (and optionally coulombic efficiency) as a function of cycle number. Multiple files from different samples can be overlaid on the same figure.

```
batplot cpc.csv --cpc –i
# single file
```

```
batplot file1.csv file2.mpt --mass 6.0 --cpc --i
# multiple files from different instrument
```

Tutorial

```
# Navigate to EC folder
batplot B443.csv B444.csv B445.csv --cpc --i
# multiple files from different instrument
```

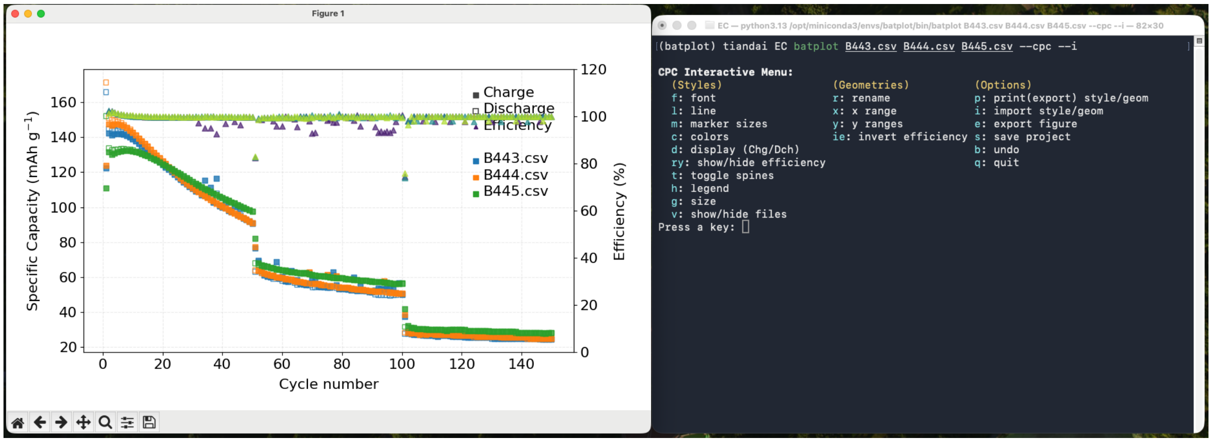

Figure 17: CPC mode

Example 6.5: Time vs. Voltage

Plot the raw time (hours) vs. voltage profile from a CSV or MPT file:

```
batplot battery.csv --xaxis time --i
```

7 Examples: *Operando* Mode

Operando mode is the most distinctive feature of *batplot*. When pointed at a folder containing sequential XRD, PDF, or XAS scans (or any text based data files) collected during an electrochemical experiment, it automatically assembles them into a 2D contour map. When a Biologic .mpt file is detected in the same folder, it also renders the GC curve as a synchronized side panel.

This dual-panel figure, correlating structural evolution with the electrochemical state of the cell, is the primary visualization type in *operando* battery research and previously required dedicated, per-experiment scripting to produce. Now, *batplot* generates it from raw files in a single command.

> **Note:** Navigate into (or specify) the folder containing the scan files before running the command. The .mpt file (if present) must be in the same folder as the scan files.

Example 7.1: Basic *Operando* Launch

Navigate to the data folder and run:

```
batplot --operando --i

# Or with explicit folder path:

batplot /path/to/operando/ --operando --i
```

batplot automatically detects all supported scan files (.xye, .qye, .xy, .dat, .txt), assembles them into a contour map using natural sorting logic, and, if a .mpt file is present, places the electrochemical voltage-time trace in a synchronized side panel.

Example 7.2: Wavelength Conversion in Operando Mode

Convert data from 2theta to Q:

```
batplot --operando --wl 0.25995 --i
```

Tutorial

```
# Navigate to Operando folder
batplot --operando -wl 1.54 --i
```

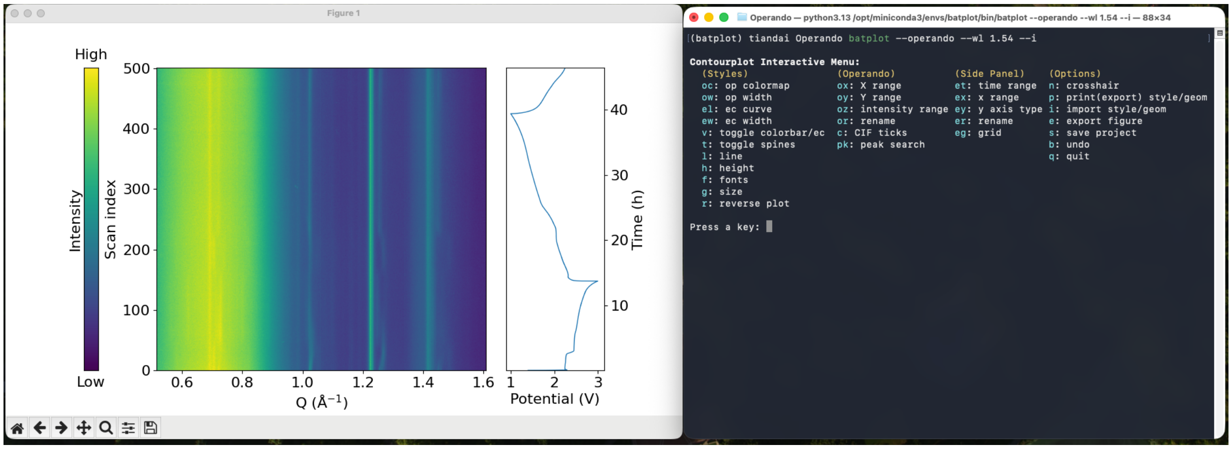

Figure 18: *Operando* mode

Example 7.3: CIF Phase Tick Marks on the Contour

CIF files can be appended to overlay Bragg reflection tick marks directly on the operando contour panel. The same wavelength conversion logic used in 1D mode applies here.

```
batplot /path/to/opereando phase1.cif phase2.cif --operando --wl 0.709 --i
```

Example 7.4: Derivative Contour

Plot the first or second derivative dy/dx of each scan as the contour, useful for XAS:

```
batplot --operando --1d --xaxis Energy --i
```

```
batplot --operando --2d --xaxis Energy --i
```

8 Examples: histogram Mode

Plot histogram based on one column of data, for example, size distribution of particles:

```
batplot --histo --i
```

batplot will ask you to specify column to plot, then min and max range and bin width before a figure is shown.

9 Batch processing

9.1 Batch Export

Export all files (same type) under the same path:

```
batplot --all --gc # Export GC plots under the path
batplot --all --xaxis 2theta # Export XRD plots under the path
```

Save all files (same type) as `pkl` sessions:

```
batplot --all --gc --save # Save GC plots as pkl session
batplot file.xy --xaxis 2theta --save
batplot file1.csv file2.csv --gc --save   # combined → name required
batplot --operando --wl 0.25 --save
batplot allfiles --xaxis q --save
batplot --all --cpc --save
batplot --all --xaxis 2theta --xrange 2 10 --save
```

9.2 Batch Edit Interactive Menu

Edit multiple saved pkl sessions for general styles and geometries (must be same type of data), this is useful when you wish to e.g. change the size of multiple plots into a same value:

```
batplot file1.pkl file2.pkl file3.pkl --i
```

10 Summary of Flags

All flags use the double-dash (--) prefix. Below are tables organized by mode.

10.1 General Flags

Flag	Description	Example
`--interactive / -i`	Open the live interactive menu alongside the figure	`batplot file.xy --interactive`
`--out FILE`	Save figure to file (default .svg if no extension)	`batplot file.xy --out plot.svg`
`--all`	Batch mode: export each file as a separate figure	`batplot --all`
`--xaxis TYPE`	X-axis type: Q, q, 2theta, r, k, energy, time, or custom	`batplot file.xy --xaxis 2theta`
`--help/--h`	Show help (add xy, ec, or op for mode-specific help)	`batplot --help ec`
`--version/--v`	Show version and release notes	`batplot --version`
`--manual/--m`	Open the illustrated manual	`batplot --manual`

10.2 1D / XY Mode Flags/Keywords

Flag	Description	Example
`--wl λ`	Wavelength (Å) for 2θ ↔ Q conversion	`batplot file.xye --wl 1.5406`
`--xrange MIN MAX`	Set X-axis display range	`batplot file.xy --xrange 10 80`
`--stack`	Stack curves vertically (auto-normalizes)	`batplot f1.xy f2.xy --stack`
`--delta N`	Spacing between stacked curves	`batplot f1.xy f2.xy --stack --delta 0.2`
`--norm`	Normalize Y intensity to the 0 to 1 range	`batplot file.xy --norm`
`--ro`	Swap X and Y axes	`batplot file.csv --xaxis time --ro`
`--1d / --2d`	Plot first derivative dy/dx	`batplot file.xy --1d --stack`
`--chik`	EXAFS χ(k) plot	`batplot data.chik --chik`
`--kchik`	EXAFS kχ(k): emphasize mid-k features	`batplot data.chik --kchik`
`--k2chik`	EXAFS k^2χ(k): most common weighting	`batplot data.chik --k2chik`

--k3chik	EXAFS $k^3\chi(k)$: emphasize high-k / heavy backscatterers	batplot data.chik --k3chik
--readcol X Y	Specify X and Y columns (1-indexed, per-file or multi-curve)	batplot file.xy --readcol 2 3
--readcol<ext> X Y	Columns for selected extension only	batplot --operando --readcoldat 2 3
--convert λ1 λ2 / q	Convert XRD and export to converted/ subfolder	batplot file.xye --convert 1.54 q batplot file.xy --convert 1.54 0.71
allfiles	Plot all files under the path	batplot allfiles --xaxis 2theta
all<ext>files	Plot all files with selected extension under the path	batplot alltxtfiles --xaxis Energy

> **Note:** allfiles is a keyword, not a flag, it is not the same as --all flag.

10.3 Electrochemistry (EC) Mode Flags

Flag	Description	Example
--gc	Galvanostatic cycling: potential vs. specific capacity	batplot file.mpt --gc --mass 7
--cv	Cyclic voltammetry: potential vs. current	batplot file.mpt --cv
--dqdv	Differential capacity: dQ/dV vs. potential	batplot file.csv --dqdv
--cpc	Capacity per cycle + coulombic efficiency	batplot file.csv --cpc
--mass MG	Active material mass in mg (required for .mpt files in GC/CPC)	batplot file.mpt --gc --mass 6.5
--xaxis time	Plot time (h) vs. potential from Neware .csv or Biologic .mpt files	batplot file.csv --xaxis time --i
--ro	Swap X and Y axes	batplot file.mpt --gc --ro --mass 7
--epc	Energy per cycle (if supported)	batplot file.csv --epc --i

10.4 *Operando* Mode Flags

Flag	Description	Example
--operando	Launch operando / contour mode from current folder or path	batplot --operando --i
--contour	Alias for --operando (identical behavior)	batplot --contour --i /path/

`--wl λ`	Wavelength (Å) to convert operando data from 2θ to Q	`batplot --operando --wl 0.25995 --i`
`--xaxis`	Specify data type	`batplot --operando --xaxis 2theta`
`--1d / --2d`	Plot derivatives of each scan as the contour	`batplot --operando --1d --i`
`--readcol`	Specify columns to plot as x and y	`Same as 1D mode`